NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Computing and Informatics
School of Computing
Department of Software Engineering

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Course Outline for PRS821S Programming for Security Professionals, Sem 1, 2026 (Honours)

Effective Start Date: Monday 9 February 2026

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect academic honesty and integrity is very important to ensure that learning is valid, reliable and credible.

NUST therefore does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their own assignments and to produce their own academic work, unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works or inventions of someone else as if it is your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

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COURSE INFORMATION

COURSE CODE AND TITLE:	PRS821S – Programming for Security Personnel
DEPARTMENT:	Software Engineering
PROGRAMMES:	08BCHS, Bachelor of Computer Science Honours (Software Development)
NATIONAL HOURS:	150 hours: (Contact: 45 hours; Directed Self-learning and Self-directed Learning: 100 hours)
NQF LEVEL AND CREDIT:	NQF Level 8, 15 Credits
SEMESTER OFFERED:	1
PRE-REQUISITES:	None
COURSE EQUIVALENCIES:	None
COURSE AIMS:	This course aims to expose students to coding techniques used to secure systems by performing hands-on projects and competing to solve challenges. The skills are appropriate for prototypes, demonstrations, and proof-of-concept code. The course will enable students to build and acquire hacking skills which will enable them to understand the techniques used by hackers to compromise systems.

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Methods of facilitating learning:
The course will be facilitated through lectures, classroom presentations, theory and practical demonstrations, discussions, assignments and exercises.

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Lectures - this will present the foundation theories of the course. Students will deepen their knowledge and understanding of the covered subjects through guided self-study.

Laboratory activities – this will be an avenue for the students to try out and acquire practical skills of theoretical knowledge gained. Laboratory materials will be prepared with clear indication of what the student is expected to achieve, problem statement and guidelines on how to approach the problem.

Course Format:

- 45 hours of face –to-face and/or remote teaching and learning time between students and lecturer
- 105 hours Directed self-learning and Self-Directed learning:
    - out-of-class study time,
    - reading prescribed material,
    - organisational overhead,
    - organising and conducting team meetings,
    - consultations with the lecturer and fellow students,
    - collecting prescribed material

Student Support and Learning Resources:

There are several facilities which students can use as a course re-source at NUST, the Faculty and at home.

NUST level: Yearbook; the library (borrow books, read short loan books and use the e-resources especially books24x7); student services especially the Writing Centre and student academic problems; Teaching & Learning Unit (TLU); Internet is available 24x7 for research.

Course level: Course Outline; Electronic learning resources on E-learning materials will be developed.

Additional support: Faculty level: school-based induction, learning support units, tutor systems
Course level: print and electronic learning resources (textbooks, modules, websites) and tutor services.

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Prescribed Textbooks:

- Khan, S. A., Kumar, R., & Khan, R. A. (2023). Software Security Concepts & Prac4ces. London: Chapman & Hall. ISBN 9781032356310

Recommended Reading:

- Seitz, J., & Arnold, T. (2021). Black Hat Python: Python Programming for Hackers and Pentesters (2nd ed.). No Starch Press. ISBN-13: 978-1718501126

- Stalling, W., & Brown, L. (2018). Computer Security Principles and Prac4ce (4th ed.). Upper Saddle River, NJ: Pearson Prentice Hall. ISBN-13: 9780134794334

- Graham, D. G. (2021). Ethical Hacking: A Hands-on Introduction to Breaking In. No Starch Press. ISBN-13-978-1718501874

Online documents:

Resources from the Internet are abundant on systems security. Referrals will be made from time to time from these resources especially pertaining to current issues on systems security such as:
https://owasp.org/(OWASP (Open Web Applica>on Security Project documents))
https://www.securecoding.org/

Quality Assurance Arrangement:

Moderation of assessments will be done in accordance with the NUST's general rules and guidelines on moderation. The programme will be re-viewed every year to cater for changes in the field. For continuous improvement, peer reviews and student evaluations will be used to provide feedback on areas of improvement.

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Lecturer Information:

Lecturer's name:	Prof Ambrose Azeta (*Course Coordinator*)
Email:	aazeta@nust.na
Office phone:	+264 813643798
Office location:	Office Building
Office Hours:	08H00 AM – 16H30 PM

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Student Readiness

Technology & Equipment Readiness:

This course will only utilize already existing NUST equipment.

Student Commitments and Contact Times:

a) **Class Attendance:** Attend all classes, and be on time, and make appropriate notes. If a student fails to attend a class, it is their responsibility to catch-up on what was covered. The students have access to the recommended books from the library. If any special presentations are used in class, it will be made available and do not need to be copied by the student.

b) **Self-Directed Study:** After every class, and before the start of the next class, sort, complete, and annotate own lecture notes, and select further reading in case any particular item was not understood on the first attempt. A student must spend their notational hours revising, practicing or/and completing homework for the course. A student should take the reading list (prescribed and recommended reading seriously). Read all prescribed material to the extent that the student could write a one-page essay, summarising the content, without looking at the material again.

c) **Communications:** Maintain active communication with your group lecturer. Students are expected to use established channels of communication and platform to solve course-related issues on time. The end of the semester is not the best time to fix issues. Use the available time to deal with matters at hand.

Online documents:

Resources from the Internet are abundant on systems security. Referrals will be made from time to time from these resources especially pertaining to current issues on systems security such as:

- https://owasp.org/(OWASP (Open Web Application Security Pro-ject documents))
- http://www.securecoding.org/

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Specific Learning Outcomes:

Upon completion of the course students will, through assessment activities, show evidence of their ability to:

- Demonstrate an in-depth understanding of secure programming practices;
- Analyse security aspects in domain-specific systems;
- Apply different techniques and tools to secure systems;
- Build secure software systems;
- Evaluate application security.

Comprehensive Learning Outcome:

Develop appropriate prototypes, demonstrations, and proof-of-concept secure code.

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Course Schedule: The following are the key knowledge areas and each knowledge area has sub-knowledge units:

WEEK #	WEEK DATES	TOPICS	Activities: Discussion Forums, Reading Materials and Assessments
Week 1	**9 – 13 FEB 2026**	**Introduction** a) Course outline and Schedule b) Learning platforms c) Problem solving concepts in computing using Python	-Set up the tone for the course -elearning platform and enrolment keys -course outline expectations based on learning outcomes. -Installation of Python.
Week 2	**16 – 20 FEB 2026**	**Introduction to python** a) Python programming language b) Structure of python program c) Writing python programs	-**Labs** (Introduction to python)
Week 3	**23 – 27 FEB 2026**	**Proactive Controls** a) Software Dependencies b) Secure Libraries	-**Labs** (Proactive Controls)
Week 4	**2 – 6 MAR 2026**	**Proactive Controls** a) Domain-based Security (Database, Web, Mobile) b) File and Memory Management Security	-**Labs** (Proactive Controls) **In-Class Exercise (Test1)**
Week 5	**9 – 13 MAR 2026**	**Secure Coding Practices** a) Network Attacks and Defence	-**Labs** (Secure Coding Practices)

Week 6	**16 – 20 MAR 2026**	**Secure Coding Practices** a) Packet Crafting and Scapy	-**Labs** (Packet Crafting and Scapy)
Week 7	**23 – 27 MAR 2026**	**Command Injection Challenges**	-**Labs** (Command Injection Challenges)
Week 8	**30 MAR – 03 APR 2026**	**MID SEMESTER BREAK**	
Week 9	**7 – 10 APR 2026**	**SQL Injection Attack and Defense**	-**Labs** (SQL Injection Attack)
Week 10	**13 – 17 APR 2026**	**Basic Networking**	-**Labs** (Basic Networking) **Comprehensive Test (Test2)**
Week 11	**20 – 24 APR 2026**	**HTTP, Authentication and Encryption**	-**Labs** (HTTP, Authentication and Encryption)
Week 12	**27 APR – 30 MAY 2026**	**Multichain and Exploit deployment**	-**Labs** (Multichain and Exploit Deployment) **Makeup/Supplementary**
Week 13	**4 – 8 MAY 2026**	**General Revision** (Consolidation of marks)	

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IMPORTANT DATES:

NOTE: The dates are subject to change based on the needs of the students at the lecturer's prerogative. Students will be notified ahead of time of any changes.

Sn	Assessment Type	Assessment Task	Date	Week
1	First Assessment	In-Class Exercise (Test1)	4 March 2026	4
2	Second Assessment	Comprehensive Test (Test2)	15 April 2026	10
3		Makeup/Supplementary	29 May 2026	12
4	Third Assessment	Laboratory Exercises (Individual Assignments)	Every week	From week 2 to week 11.
5		Dates for Consolidation of Semester marks	4 to 8 May 2026	13
6	Examination		TBA	

ASSESSMENT STRATEGIES:

The course will be assessed using a combination of Continuous Assessment and an end-of-semester examination. Continuous Assessment is made up of the following: Tests, Assignments/Projects, Exercises, Quizzes, and Tutorials.

Students have to obtain at least 40% to write the examination while a sub-minimum of 40% is required in the examination. The semester mark and examination mark shall be used jointly to determine the final mark in the ratio of 40% (semester mark) to 60% (examination mark). A minimum final mark of 50% is required to pass the course. Total Course Mark: 100%

	Continuous Assessment (Semester Mark)	Weight
	Test 1	35%
	Test 2	35%
	Laboratory Exercises (Individual Assignments)	30%
A.	Total Semester Mark 100% (minimum 40% required to write examination)	40%
B.	Assessment (Examination Mark 60%)	60%
	Total Final Course Mark (A + B)	100%

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Plagiarism and Deduction of Marks:

Marks will be deducted if students do not adhere to the rules of the University according to Rule AC3.2. All assignments should be submitted through Turnitin, the similarity software that is integrated into the MOODLE Learning Management System. If plagiarism is detected, marks should be deducted as follows:

It is the student's responsibility to be familiar with and adhere to the NUST's Policies. These Policies can be found in the NUST Prospectus or online at:

https://www.nust.na/sites/default/files/documents/eYearbook_P4_ComputingandInformatics.pdf

% Similarity Detected	% Marks Deduction
0 – 20	0
20 – 40	10
40 – 60	25
60 – 100	100

**** Please note that the % shown above is an average % for all Faculties and should be used as a Guideline.***

Assignments found with a similarity report above 20%, will not be allowed to apply for a remark or a recheck of marks. For students who fall into the 60 to 100% similarity group, Rule AC3.2 will apply, and the misconduct procedure will start.

Kindly note the difference between similarity and plagiarism.
The Turnitin software detects the similarity and assists staff and students with academic integrity. It should be used as follow:

- Pay attention to the repository setting. It should be set to "no repository" to avoid self plagiarism. Please note that the submitted work is compared with existing information in the database and other work that is available on the internet, such as electronic journals, websites, and internet resources.
- Turnitin should not be seen as a system that detects plagiarism, but it generates a similarity report. It identifies text matches and therefore guides staff and students to pay attention t o certain areas. A high text match does not indicate plagiarism, but rather a match with existing content in the database.
- A low level match, on the other hand might be plagiarised work which Turnitin could not detect because it is from a source that cannot be accessed or not available in the database.
- Set the error margin not to detect references.

Plagiarism is the act of taking someone else's work or ideas and passing them off as one's own and by failing to include quotations or give the appropriate citation by not adequately acknowledging an author of a source (NUST Policy on Plagiarism, 2020).
Lecturers will assess if the detected similarities are plagiarized materials
If it is found that the materials are plagiarised, then action as indicated in Rule AC3.2 will be taken.

Assessment Submission:
All assessments (assignments, projects, and activities) should be submitted on the eLearning platform to ensure that it goes through the Turnitin software to check for similarity.

Assessments should be typed and submitted in MS Word or PDF format. Please note that students should not submit an assignment as a JPG or any other image format, because Turnitin cannot read images.

Course Policies

General Academic Policies:

Failure to Pay Fees:
A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

Important Student Services at NUST
There are a variety of services that you can use at NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development - Dean of Students Office.
- The Writing Centre and student academic problems – Centre for Teaching & Learning (CTL)
- Campus Health and Wellness Centre (CHWC) - Dean of Student's office/ NUST Clinic

AUTHORISATION:

This course is authorised for use by:

________________________ ______________________________

Head of Department Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I, _____________________________________, hereby acknowledge that I have received this course outline for (insert course title and code), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

_________________________ ______________________________

Signature of Student Date